

Skills Summary

Completing Short Algebraic and Segment Proofs

Use this reference while completing your worksheet or when studying.

Skill 1 — Naming the property behind each step

Properties of equality — one per line of an algebraic proof:

- **Addition / Subtraction:** add or subtract the same quantity on both sides.
- **Multiplication / Division:** multiply or divide both sides by the same non-zero quantity.
- **Distributive:** $a(b + c) = ab + ac$ — rewriting, not operating on both sides.
- **Substitution:** replace a quantity by one known to equal it.

Name the *property*, not the arithmetic: “divided by 3” is what you did; “division property of equality” is why you were allowed to.

Example: Prove $x = 8$ from $4x - 9 = 23$, naming a reason for each line.

Step 1. Each line of the proof undoes one operation, so I choose the property that names the operation I am undoing.

Step 2. $4x - 9 = 23$ (given); $4x = 32$ (addition property of equality); then divide both sides by 4 (division property of equality).

$$\Rightarrow x = 8$$

Try it: Prove a value of x from $6x + 5 = 41$, naming a property for each line.

$$9 = x \text{ :check}$$

Skill 2 — Segment addition as an equation

Segment Addition Postulate: if B is between A and C , then $AB + BC = AC$.

Definition of midpoint: M on $\overline{AB}$ with $\overline{AM} \cong \overline{MB}$.

Definition of congruent segments: $\overline{AM} \cong \overline{MB}$ means $AM = MB$ — the step that turns a congruence mark into an equation you can solve.

The order matters in a proof: betweenness gives the sum; the definitions give the equalities.

Example: B is between A and C , with $AB = 3x$ m, $BC = x + 2$ m and $AC = 26$ m. Find x .

Step 1. The stated betweenness is exactly the hypothesis of the Segment Addition Postulate, so the picture becomes the one equation $AB + BC = AC$.

Step 2. $3x + (x + 2) = 26$, so $4x + 2 = 26$ and $4x = 24$.

$$\Rightarrow x = 6$$

Try it: B is between A and C , with $AB = 2x$ m, $BC = x + 5$ m, $AC = 29$ m. Find x .

$$8 = x \text{ :check}$$

Skill 3 — Overlapping segments in a two-column proof

For four points in order, the two *overlapping* distances share a middle stretch.

The move: add the shared middle stretch to both sides of the given equality (addition property of equality), then rename each sum with the Segment Addition Postulate, then substitute.

Running it backwards — *subtracting* the shared stretch — takes you from equal overlapping distances to equal outer stretches.

Example: Points A, B, C, D lie in order with the outer stretches equal, $AB = CD = a$ cm and the middle stretch $BC = b$ cm. With $a = 5$ and $b = 8$, find the overlapping distance AC .

Step 1. Adding the shared middle stretch to each equal outer stretch keeps the two sides equal, and the Segment Addition Postulate names those sums AC and BD — so computing one of them computes both.

Step 2. $AC = a + b = 5 + 8$, and $BD = b + a$ is the same total.

$\Rightarrow$ **13** cm, and BD equals it.

Try it: Same picture with $a = 6.5$ cm and $b = 9$ cm. Find AC .

check: 15.5